Roll No. | | | | | | | |

MID SEMESTER EXAMINATION SEPTEMBER- 2025

Name of the Course: B. Tech — Semester: I

Name of the Paper: Engineering Mathematics for AI-I — Paper Code: TMA 102

Time: 90 Minutes — Maximum Marks: 50

Note:-

(i) Answer all the questions by choosing any one of the sub questions.

(ii) Each question carries 10 marks.

Q1		(10 marks)
(a)	Determine the values of b such that the rank of matrix A is 3: $A=\begin{bmatrix}1&1&-1&0\\4&4&-3&1\\b&2&2&2\\9&9&b&3\end{bmatrix}$.	(CO1)
	OR	
(b)	Find matrix P such that $P^{-1}AP$ is a diagonal matrix, where $A=\begin{bmatrix}1&6&1\\1&2&0\\0&0&3\end{bmatrix}$.	(CO1)
Q2		(10 marks)
(a)	Examine the following vectors for linear dependence and find the relation if it exists. $X_1=(1,3,4,2)$, $X_2=(3,-5,2,2)$, $X_3=(2,-1,3,2)$.	(CO1)
	OR	
(b)	Evaluate the following limit $\lim_{x\to 0}\left(\frac{1}{x}\right)^{\tan x}$	(CO3)
Q3		(10 marks)
(a)	Find the eigen values and the corresponding eigen vectors for the following matrix. $A=\begin{bmatrix}3&10&5\\-2&-3&-4\\3&5&7\end{bmatrix}$.	(CO1)
	OR	
(b)	If $u=log(x^3+y^3+z^3-3xyz)$ then show that: $\left(\frac{\partial}{\partial x}+\frac{\partial}{\partial y}+\frac{\partial}{\partial z}\right)^2 u=-\frac{9}{(x+y+z)^2}$.	(CO3)
Q4		(10 marks)
(a)	Verify the Cayley Hamilton Theorem for the following matrix. Hence, find A^{-1}. $A=\begin{bmatrix}1&4\\2&3\end{bmatrix}$ Also, find $B=A^5-4A^4-7A^3+11A^2-A-10I$	(CO1)
	OR	
(b)	If $y=e^{m\sin^{-1}x}$, prove that $(1-x^2)y_{n+2}-(2n+1)xy_{n+1}-(m^2+n^2)y_n=0$.	(CO3)
Q5		(10 marks)
(a)	Find a Single Value Decomposition for $A=\begin{bmatrix}1&0&1\\-1&1&0\end{bmatrix}$.	(CO1)
	OR	
(b)	Find y_n, if $y=e^{2x}.cosx.sin^2 2x$.	(CO3)